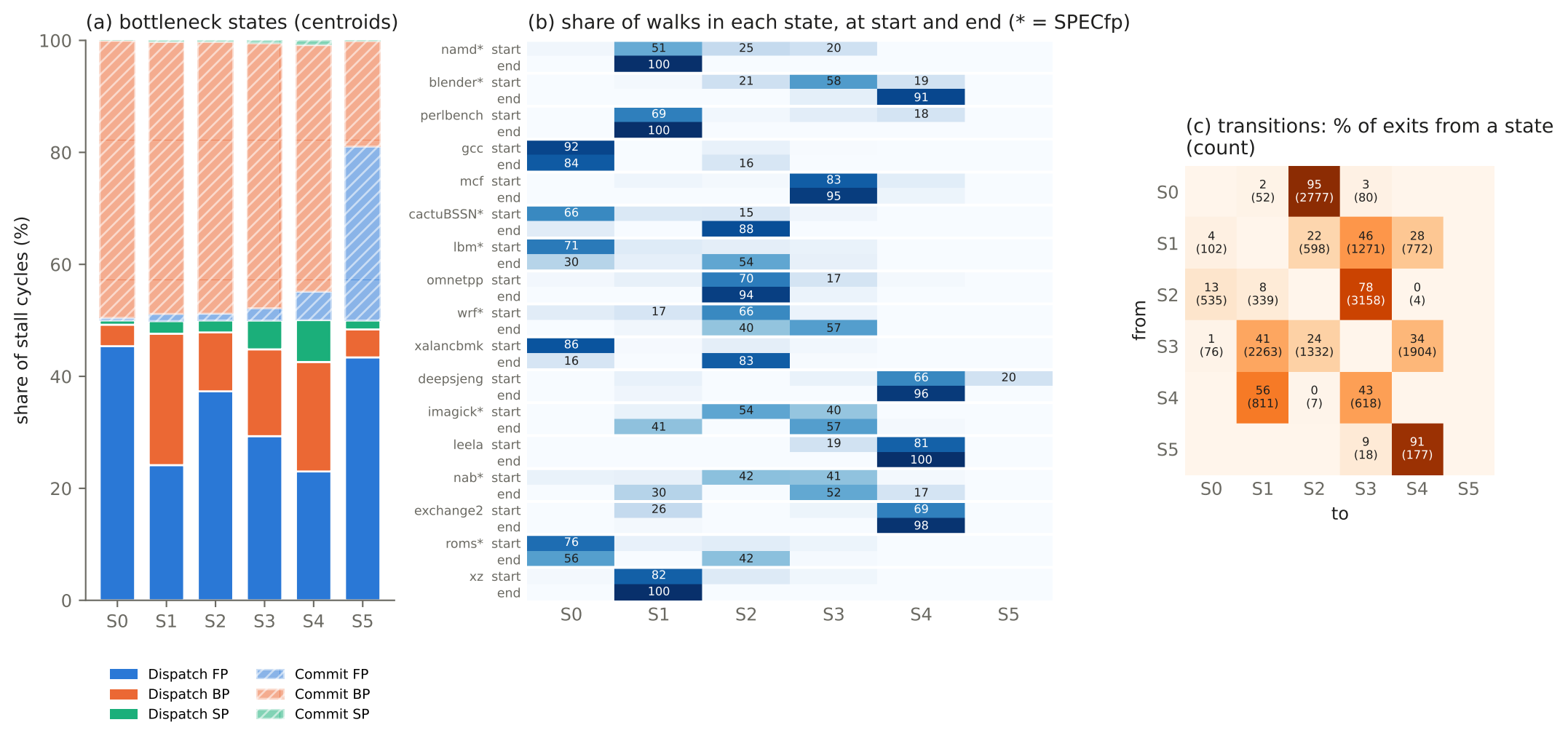
(a) bottleneck states (centroids)
share of stall cycles (%)
100
80
60
40
20
0
S0
S1
S2
S3
S4
S5
Dispatch FP
Dispatch BP
Dispatch SP
Commit FP
Commit BP
Commit SP
(b) share of walks in each state, at start and end (* = SPECfp)
namd* start 51 25 20
end 100
blender* start 21 58 19
end 91
perlbench start 69 18
end 100
gcc start 92
end 84 16
mcf start 83
end 95
cactuBSSN* start 66 15
end 88
lbm* start 71
end 30 54
omnetpp start 70 17
end 94
wrf* start 17 66
end 40 57
xalancbmk start 86
end 16 83
deepsjeng start 66 20
end 96
imagick* start 54 40
end 41 57
leela start 19 81
end 100
nab* start 42 41
end 30 52 17
exchange2 start 26 69
end 98
roms* start 76
end 56 42
xz start 82
end 100
S0
S1
S2
S3
S4
S5
(c) transitions: % of exits from a state (count)
from
S0
2 (52)
95 (2777)
3 (80)
S1
4 (102)
22 (598)
46 (1271)
28 (772)
S2
13 (535)
8 (339)
78 (3158)
0 (4)
S3
1 (76)
41 (2263)
24 (1332)
34 (1904)
S4
56 (811)
0 (7)
43 (618)
S5
9 (18)
91 (177)
S0
S1
S2
S3
S4
S5
to